

46 YEARS OF EXCELLENCE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-15** **Date: 09.01.2025**

JEE MAIN MODEL

MATHS

GIVEN QUESTIONS

51. Let $P(\alpha,\beta,\gamma)$ be the image of the point $Q(3, -3, A)$ in the line $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}$ and R be the point $(2, 5, -A)$. If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to:

A) 81 B) 36 C) 18 D) 72

56. Let $\vec{a}=6\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a is vector such that $|\vec{c}|\geq 6$, $\vec{a}.\vec{c}=6|\vec{c}|$, $|\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 60°, then $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ is equal to:

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A) $\frac{3}{2}\sqrt{6}$ B) $\frac{9}{2}\left(6+\sqrt{6}\right)$ C) $\frac{9}{2}\left(6-\sqrt{6}\right)$ D) $\frac{3}{2}\sqrt{3}$

66. If the locus of the point, whose distance from the point (2,A) and (1,C) are in the ratio 5:4, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to:

A) 37 B) 437 C) 5 D) -27

73. From a lot of 12 items containing 3 defective, a sample of 5 items is drawn at random. Let the random variable X denote the number or defective items in the sample. Let items in the sample be drawn one by one without replacement. If variance of X is $\frac{m}{n}$, where gcd(m, n) = 1, then $n - m$ is equal to ___

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CORRECT QUESTIONS.

51. Let $P(\alpha,\beta,\gamma)$ be the image of the point $Q(3,-3,1)$ in the line $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}$ and R be the point $(2,5,-1)$. If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to:

A) 81 B) 36 C) 18 D) 72

56. Let $\vec{a}=6\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a vector such that $|\vec{c}|\geq 6$, $\vec{a}.\vec{c}=6|\vec{c}|$, $|\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 60°, then $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ is equal to:

A) $\frac{3}{2}\sqrt{6}$ B) $\frac{9}{2}\left(6+\sqrt{6}\right)$ C) $\frac{9}{2}\left(6-\sqrt{6}\right)$ D) $\frac{3}{2}\sqrt{3}$

66. If the locus of the point, whose distance from the point $(2,1)$ and $(1,3)$ are in the ratio 5:4, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to:

A) 37 B) 437 C) 5 D) -27

73. From a lot of 12 items containing 3 defective, a sample of 5 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Let items in the sample be drawn one by one without replacement. If variance of X is $\frac{m}{n}$, where gcd(m, n) = 1, then $n - m$ is equal to ___

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